

MediCrypt – Secure Medical Consultation & Health Record Management System

1. Introduction

The database is the core component of the MediCrypt system. It securely stores information related to users, appointments, consultations, prescriptions, medical records, and system activities. A well-designed database helps maintain data integrity, improves performance, and ensures patient information remains secure.

2. Database Name

Database Name: medicrypt_db

This database is used to manage all the information required for the MediCrypt application.

3. Database Tables

The system contains the following main tables.

Table Name	Purpose
Users	Stores Admin, Doctor, and Patient information.
Appointments	Stores appointment booking details.
Consultations	Stores doctor's consultation notes securely.
Medical_Records	Stores uploaded medical reports.
Prescriptions	Stores medicines prescribed by doctors.
Audit_Logs	Stores important user activities for security and tracking.

4. Users Table Overview

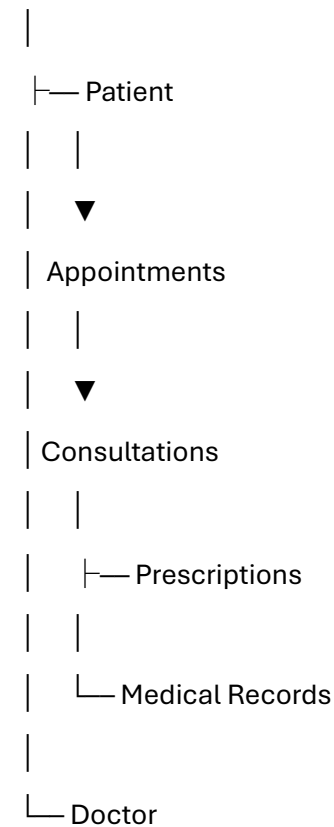
Instead of creating separate tables for Admin, Doctor, and Patient, the system uses a single **Users** table. The **Role** column identifies the type of user.

Column	Description
user_id	Unique user ID
full_name	Full name of the user
email	Email address
password	Hashed password
role	Admin / Doctor / Patient

Column	Description
phone	Contact number
status	Active or Inactive
created_at	Account creation date

5. Database Relationships

Users



6. Security Planning

To protect sensitive medical information, the system follows the following security practices.

Password Security

- Passwords are stored using hashing.
- Original passwords are never stored in the database.

Encrypted Data

The following information will be encrypted:

- Consultation Notes

- Medical Reports
- Prescriptions

Normal Data

The following information is stored without encryption:

- Name
- Email
- Phone Number
- Appointment Date